

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Eight Semester of B. Tech. Examination (EE)

April 2013

EE 406 Electrical Machine Design-II

Date: 22.04.2013, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Explain parameters that must be kept in mind while selecting the number of stator slot during the design of three phase induction motor. [07]

Q - 2 (a) Discuss the factors affecting selection of air gap of three phase induction motor. Give different formula for it. [07]

Q - 2 (b) During the stator design of a 3 phase, 30 kW, 400 V, 6 pole, 50 Hz, squirrel cage induction motor, following information has been obtained. Gross length of stator = 0.17 m, Internal diameter of stator = 0.33 m, Number of stator slots = 45, Number of conductors per slot = 12. Based on the above data, design a suitable cage rotor for the above motor. [07]

OR

Q - 2 (a) What is dispersion coefficient? Explain the effect of dispersion coefficient on power factor of induction motor. [07]

Q - 2 (b) Estimate the stator core dimensions, number of stator slots and number of stator conductors per slot for a 100kW, 3300V, 50Hz, 12 pole star connected slip ring induction motor. Assume average flux density=0.4wb/m², ampere conductors per metre=25000, efficiency=0.9 and power factor=0.9. Choose the main dimensions to give best power factor. Assume winding factor=0.96. The slot loading should not exceed 500 ampere conductor. [07]

Q - 3 (a) Explain briefly harmonic torques and in brief harmonic reduction techniques. [07]

Q - 3 (b) During the design of a 3 phase, 5 kW, 400 V, 50 Hz, 4 pole, squirrel cage induction motor designed for star delta starting, following data have been obtained. Gross length of stator core = 0.11 m, Rotor diameter = 0.14 m, Turns per phase on stator = 360, Air gap length = 0.45 mm, Carter's gap contraction co efficient = 1.25. Assuming that the ampere turns required for the iron parts are about 32 percent of that needed for the gap, calculate the magnetizing current drawn by the motor and comment upon its value. [07]

OR

Q - 3 (b) Derive the expression of rotor bar and end ring current in case of 3 phase squirrel cage induction motor. [07]

SECTION – II

- Q - 4 Explain the term Short Circuit Ratio and its effect on synchronous machines performance. [07]
- Q - 5 (a) Discuss briefly the factors which influence the selection of the specific magnetic loading and specific electric loadings in the design of synchronous machines. [07]
- Q - 5 (b) Estimate suitable air gap diameter and length of a 10 MVA, 11 kV, 8-pole 3-phase, 50 Hz star connected synchronous generator. Maximum air gap flux density is 0.92 Tesla. The ampere per conductor per metre length of periphery is varying from 20,000 to 40,000 Ac/m. The peripheral velocity should not be more than 80 m/sec. Suggest the type of pole to be constructed. Pole arc to pole pitch ratio is 0.66. [07]

OR

- Q - 5 (a) Explain the procedure for design of the damper winding of a 3 phase salient pole generator. [07]
- Q - 5 (b) Following information have been obtained during the design of stator frame of a 500 kVA, 6.6 kV, 12 pole 500 rpm, star connected, 3 phase salient pole alternator. Stator diameter at gap surface = 1.3 m, Gross length of stator core = 0.21 m, Gap flux per pole = 0.0404 Weber. Based upon these information's, design the field system of this alternator, giving the following details. (i) Length of air gap (ii) Diameter of rotor at gap surface (iii) Peripheral speed (iv) Length, width and height of the pole (v) Depth of the rotor core. Assume suitable data where ever necessary. [07]
- Q - 6 (a) Determine the ampere turns required by the pole of an engine driven salient pole alternator which has the following specifications and dimensions: 1875 kVA, 3.3 kV, 50 Hz, 3 phase, 250 rpm. Internal diameter of stator = 238 cm, Gross core length = 38 cm, Flux per pole = 0.068 Weber, Length of air gap = 0.6 cm, Height of the pole = 15 cm, Width of the pole = 16 cm. Ampere turns for the stator teeth and stator core taken together may be assumed 15 % of the air gap ampere turns. [07]
- Q - 6 (b) A 1250 kVA, 3 phase, 6600 V, salient pole alternator has the following data: Air gap diameter 1.6 m, length of core 0.45 m, number of poles 20, armature ampere conductors per metre 28000, ratio of pole arc to pole pitch 0.68, stator slot pitch 28 mm, current density in damper bars 3 A/mm². Design a suitable damper winding for the machine. [07]

OR

- Q - 6 (b) Draw and explain the flow chart for overall design of synchronous machine [07]
